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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. (EC) (C.S.E.) Sem. - I Examination, 2009-10
EC – 704 Digital Signal Processing & Applications

Date: 28/12/2009, Monday

Time: 1:00 p.m to 4:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION -I

Q-1(a) The analog signal given below is sampled by 600 samples per second 3

$x(t) = 2\sin 480\pi t + 3\sin 720\pi t$, Calculate:

- 1) Nyquist sampling rate and maximum frequency
- 2) What are the frequencies in the radians in the resulting discrete signal $x(n)$?

(b) Give Difference equation representation of Moving-Average system. 4
Find the frequency response of the Moving-Average system.

(c) Determine Impulse response for a stable linear time-invariant system 3
for which the input $x[n]$ and output $y[n]$ satisfy the linear constant-coefficient difference equation

$$y[n] - 0.5y[n-1] = x[n] - 0.25x[n-1]$$

(d) Sketch the R.O.C. of delayed unit impulse sequence $\delta[n-7]$. 1

Q-2(a) Consider the three sequences 5

$$v[n] = u[n] - u[n-6],$$

$$w[n] = \delta[n] + 2\delta[n-2] + \delta[n-4],$$

$$q[n] = v[n] * w[n]$$

(a) Find and sketch the sequence $q[n]$.

(b) Find and sketch the sequence $r[n]$ such that $r[n] * v[n] = \sum_{k=-\infty}^{n-1} q[k]$

- (b) Give decomposition of a stable, causal system into the cascade of a minimum-phase and an all-pass system; consider the two stable, causal systems specified by the system functions. 4

$$H_1(z) = \frac{(1+3z^{-1})}{(1+\frac{1}{2}z^{-1})} \quad H_2(z) = \frac{(1+\frac{3}{2}e^{j\pi/4}z^{-1})(1+\frac{3}{2}e^{-j\pi/4}z^{-1})}{(1-\frac{1}{3}z^{-1})}$$

- (c) Explain All-Pass systems with its typical pole-zero plot 3

OR

- Q-2(a) State whether the following systems are Linear, Time-invariant, Causal, Stable 4

$$T(x[n]) = y[n] = (\cos \pi n)x[n].$$

- (b) The system function of a causal linear time-invariant system is 4

$$H(z) = \frac{1-z^{-1}}{1+\frac{3}{4}z^{-1}},$$

The input to this system is

$$x[n] = (1/3)^n u[n] + u[-n-1].$$

- (a) Find the impulse response of the system $h[n]$.
 (b) Find the output $y[n]$.
 (c) Explain properties of R.O.C. for the z-transform. 4
- Q-3(a) Draw the structures of cascade and parallel realizations of 4

$$H(z) = \frac{(1-z^{-1})^3}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{8}z^{-1})}$$

- (b) Explain Reconstruction of Bandlimited signal from its samples. 4

- (c) Determine the z-transform and sketch the R.O.C. for following sequence. $x[n] = (-1/3)^n u[n] - (1/2)^n u[-n-1]$.

4

OR

- Q-3(a) Draw direct form I and II realization of following transfer function

$$H(z) = \frac{(1 - \frac{1}{2}z^{-2})}{(1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2})}$$

4

- (b) Determine the inverse z-transform of

4

$$X(z) = \frac{z}{3z^2 - 4z + 1}$$

If the regions of convergence are (a) $|z| > 1$, (b) $|z| < 1/3$ and

(c) $1/3 < |z| < 1$

- (c) Compute convolution of following sequences.

4

$$x(n) = n \text{ for } 0 \leq n \leq 4$$

$$h(n) = 1 \text{ for } -2 \leq n \leq 2$$

0 elsewhere.

0 elsewhere

SECTION -II

- Q-4(a) Draw and explain a radix-2 Decimation in time FFT implementation signal flow graph (butterfly diagram) for 8 point FFT.

7

- (b) Discuss the following properties of DFT

4

(i) Linearity (ii) Duality.

- Q-5(a) Compute the DFTs of the sequence $x[n] = \cos n\pi/2$ where $N=4$, using Decimation in frequency FFT algorithm

4

- (b) Find the inverse DFT of $X(k) = \{1, 2, 3, 4\}$.

4

- (c) Explain polyphase decomposition in brief.

4

OR

Q-5(a) Design second order butterworth band pass filter that meet following specifications: 7

Lower cutoff frequency = 210 Hz

Upper cutoff frequency = 330 Hz

Sampling frequency = 960 Sp/s

(b) Explain bilinear transformation in brief. 5

Q-6(a) Explain filter design by impulse invariance method in detail. 7

(b) Explain Discrete Wavelet transform in detail. 5

OR

Q-6(a) Explain how 2D signal processing is helpful in processing the images 7

(b) Compare FIR and IIR filters. 5
